Real-Time Driver Drowsiness Detection

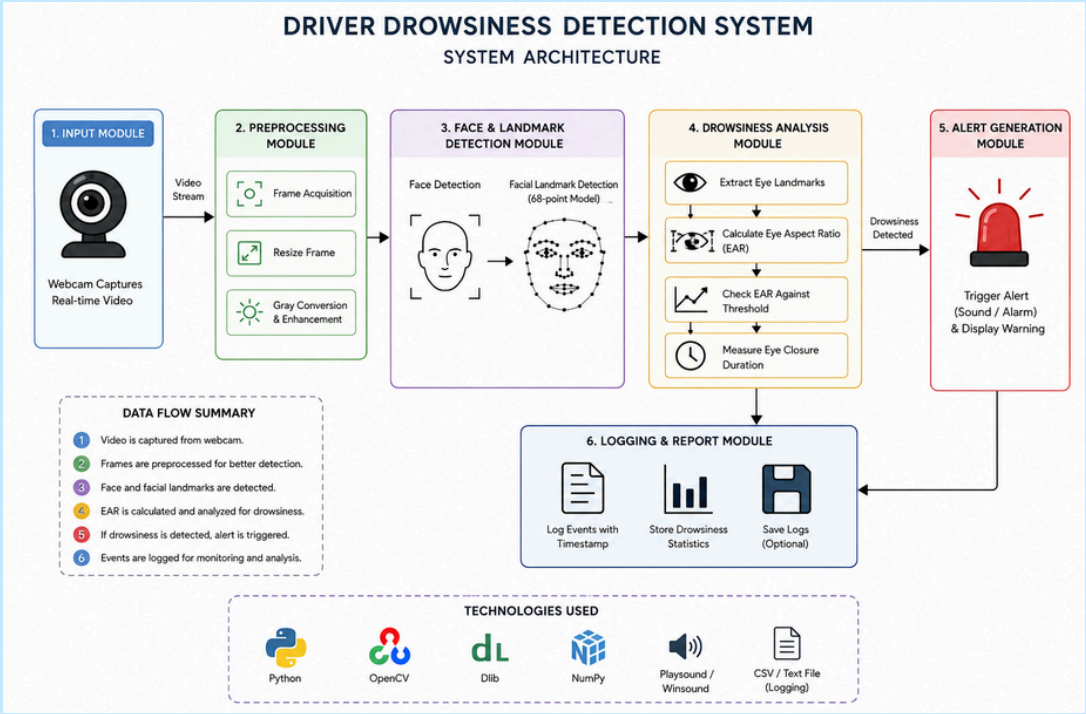
An Automated Fatigue Monitoring System Utilizing Python, OpenCV, and Dlib Facial Landmark Detection

Introduction

- Driver drowsiness is one of the leading causes of road accidents worldwide, resulting in serious injuries and loss of life.
- To address this issue, a real-time driver drowsiness detection system has been developed using computer vision and machine learning techniques.
- The system continuously monitors the driver's eye movements and facial features through a camera.
- It uses the Eye Aspect Ratio (EAR) method to accurately detect signs of fatigue and drowsiness.

Objectives:

- **To develop a real-time system for detecting driver drowsiness**
- **To monitor driver's eye movements and facial features using a camera and use EAR**
- **To analyze eye closure duration to identify signs of sleepiness and detect it**
- **To provide instant visual and audible alerts to the driver**
- **To improve overall road safety through automation**



Limitations:

- Reliable performance requires adequate lighting; accuracy may decrease in poor or low-light environments.
- The detection of facial landmarks can be hindered by objects like sunglasses.
- Extreme or sudden head movements can disrupt the tracking and analysis of facial features.

Outcoms:

- Accurately detects drowsiness using Eye Aspect Ratio (EAR)
- Tracks eye closure duration to measure fatigue level
- Provides instant visual and audible alerts when drowsiness is detected
- Cost-effective and suitable for real-world implementation
- Improves overall driving safety

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